

CISCO Packet Tracer Lab 3 Report

Network Monitoring & Security Analysis

Instructor:

Ramy Othman

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Team: Emily Nunez

Angelee Sullivan-Quintana

Isabella Zárate González

Kyrellos Tawadros

Lab Objective

The objective of this lab was to simulate and analyze network traffic using Cisco Packet Tracer's Simulation Mode as a substitute for Wireshark. We built a basic network with an ISR4331 Router, a 2960-24TT Switch, three PCs (a monitoring PC, a victim PC, and an attacker PC), and a Server. The lab focused on capturing packet details including IP addresses, MAC addresses, and protocols, simulating a ping flood attack, detecting port scanning by probing open services on a server using Telnet, and identifying traffic anomalies caused by high volume simultaneous traffic from multiple sources.

Tasks:

1. Capture network traffic and analyze IP addresses, MAC addresses, and protocols.
2. Simulate a ping flood attack and observe network behavior.
3. Detect port scanning attempts using Telnet to probe open ports on a server.
4. Identify traffic anomalies such as high volume and unusual packet behavior.

Equipment and Resources

Hardware:

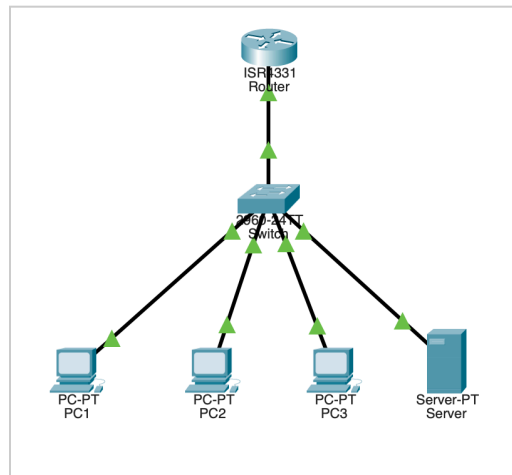
- 1x Cisco ISR4331 Router
- 1x Cisco 2960-24TT Switch
- 3x PC End Devices (PC1 - Monitoring, PC2 - Victim, PC3 - Attacker)
- 1x Server-PT (target for port scanning)
- 4x Copper Straight-Through Cables
- Cisco Packet Tracer (Network Simulation Software)

Software:

- Cisco Packet Tracer (Network Simulation Software)

Lab Steps

Step 1: Network Topology



We built the network topology by placing one ISR4331 Router, one 2960-24TT Switch, three PC end devices, and one Server-PT onto the workspace. All devices were connected using Copper Straight-Through cables. PC1 was designated the Monitoring PC, PC2 the Victim PC, and PC3 the Attacker PC. The Server was assigned IP address 192.168.1.50. The router interface GigabitEthernet0/0/0 was configured with IP 192.168.1.1 and brought up using the no shutdown command.

Step 2: Router Configuration

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

e. Delivery of Cisco cryptographic products does not imply
importer, exporter, distributor and users are responsible for
compliance with U.S. and local country laws. By using this product you
are to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wel/export/crypto/tech/stgsp.html

you require further assistance please contact us by sending email to
support@cisco.com.

sco ISR4331/K9 (1RU) processor with 1795999K/4147K bytes of memory.
processor board ID F4M23201000
Gigabit Ethernet interfaces
768K bytes of non-volatile configuration memory.
94304K bytes of physical memory.
57167K bytes of flash memory at bootflash.
bytes of WebUI CDM files at webui.

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

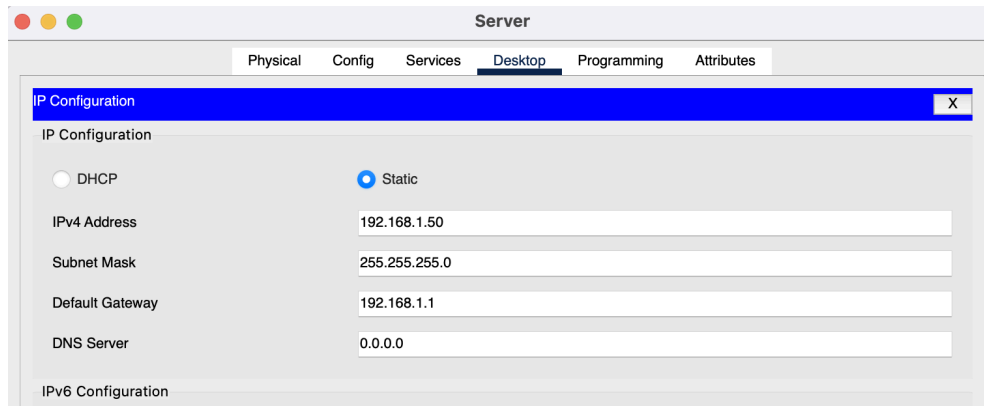
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Please specify interface type and number
Router(config)#interface GigabitEthernet0/0
Please specify interface type and number
Router(config)#do show ip interface brief
Interface IP-Address MTU Method Status Protocol
GigabitEthernet0/0/0 unassigned YES unset administratively down down
GigabitEthernet0/0/1 unassigned YES unset administratively down down
GigabitEthernet0/0/2 unassigned YES unset administratively down down
GigabitEthernet0/0/3 unassigned YES unset administratively down down
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Interface GigabitEthernet0/0/0, changed state to up
Interface GigabitEthernet0/0/0, line protocol on Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#exit
Router#
Router#
```

We configured the router through the CLI. After running show ip interface brief, we identified that the correct interface was GigabitEthernet0/0/0. We assigned it the IP address 192.168.1.1 with subnet mask 255.255.255.0 and brought the interface up using the no shutdown command.

Step 3: PC IP Configuration

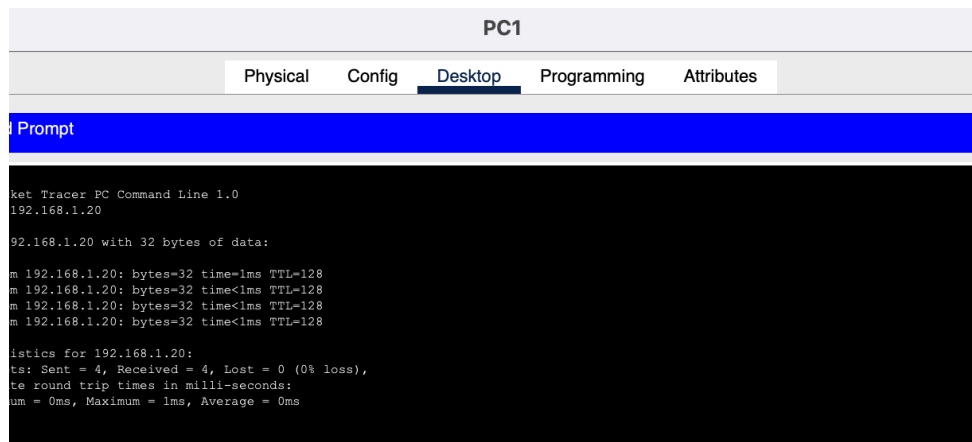
The image displays three screenshots of PC configuration windows, each showing the 'IP Configuration' tab. The windows are titled PC1, PC2, and PC3. Each window has tabs for Physical, Config, Desktop, Programming, and Attributes. The 'IP Configuration' section shows the following details:

PC	Interface	IP Configuration Type	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC1	FastEthernet0	Static	192.168.1.10	255.255.255.0	192.168.1.1	0.0.0.0
PC2	FastEthernet0	Static	192.168.1.20	255.255.255.0	192.168.1.1	0.0.0.0
PC3	FastEthernet0	Static	192.168.1.30	255.255.255.0	192.168.1.1	0.0.0.0



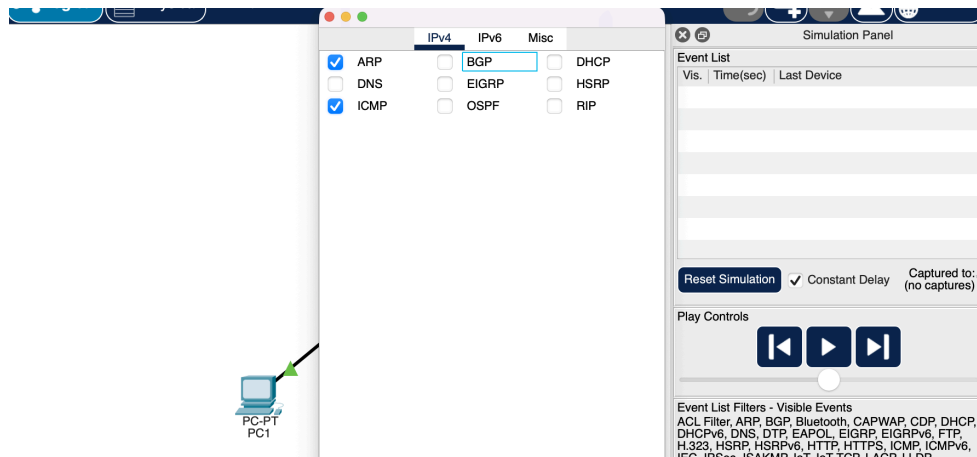
Each PC was statically configured with an IP address. PC1 was assigned 192.168.1.10, PC2 was assigned 192.168.1.20, and PC3 was assigned 192.168.1.30. All devices used subnet mask 255.255.255.0 and default gateway 192.168.1.1.

Step 4: Network Connectivity Test



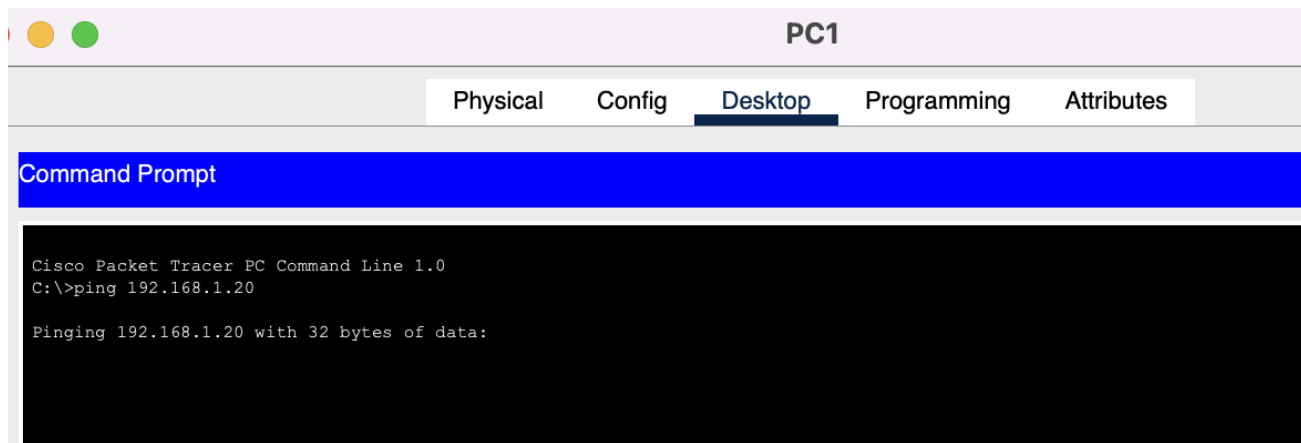
To verify the network was functioning correctly, we pinged 192.168.1.20 from PC1. We received 4 replies with 0% packet loss, confirming all devices could communicate across the network before beginning the simulation tasks.

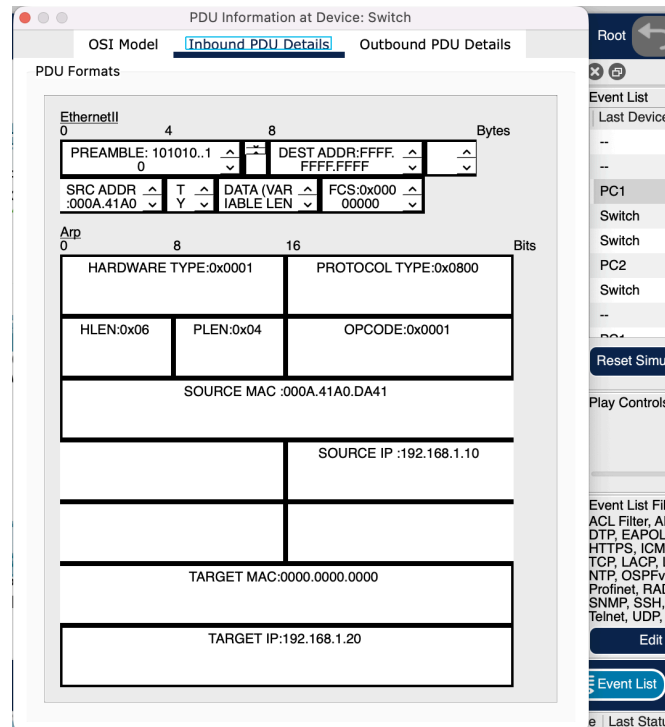
Step 5: Switch to Simulation Mode and Set Filters



We switched to Simulation Mode and clicked Edit Filters. On the IPv4 tab, we checked only ARP and ICMP. This allowed us to capture the initial ARP resolution and ICMP ping traffic without noise from other background protocols.

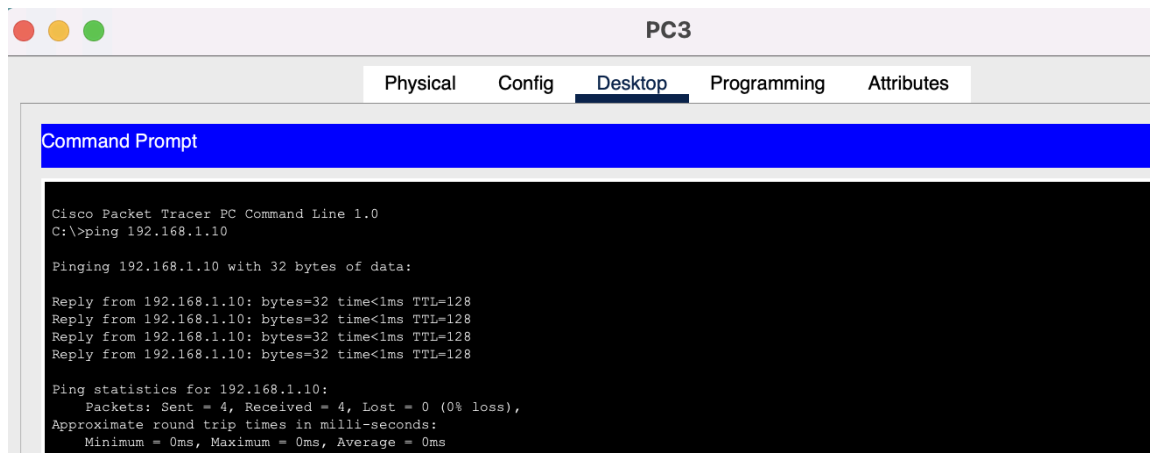
Step 6: Capture a Ping and View Packet Details

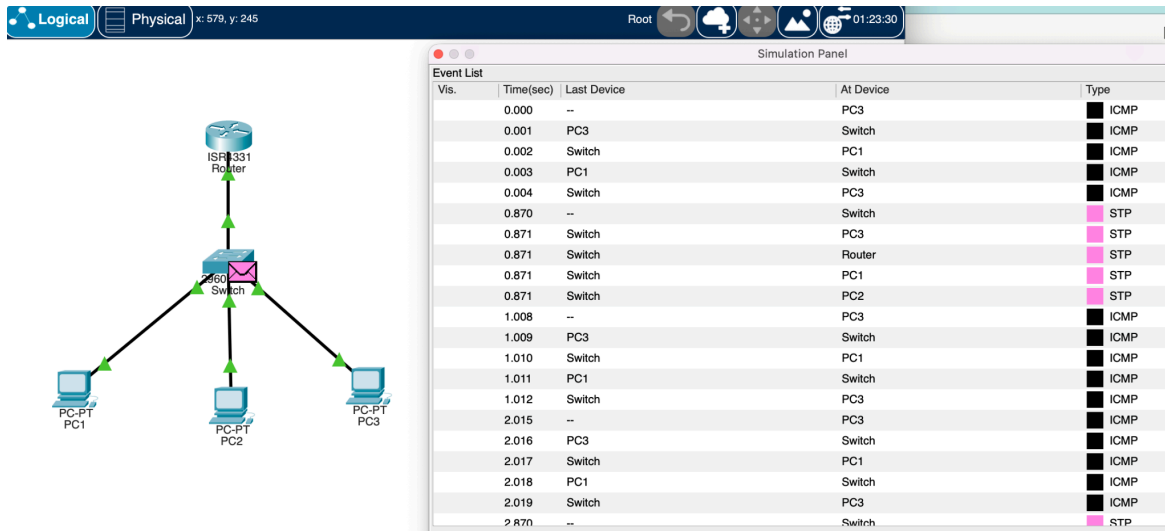




We ran a ping from PC1 to PC2 in Simulation Mode and opened the PDU Information window to select the Inbound PDU Details tab. The packet details showed the source MAC address, the ARP protocol information including the Source IP and Target IP, and confirmed the protocol type as ARP (0x0806). This captured the key network identifiers of IP address, MAC address, and protocol.

Step 7: Create Ping Flood from Attacker PC





We reset the simulation and updated the filters to show ICMP only. From PC3, we ran a ping targeting PC1. The Event List rapidly filled with ICMP packets originating from PC3 bouncing to the Switch and then to PC1, simulating the behavior of a ping flood denial-of-service attack.

Step 8: Verify Open Ports

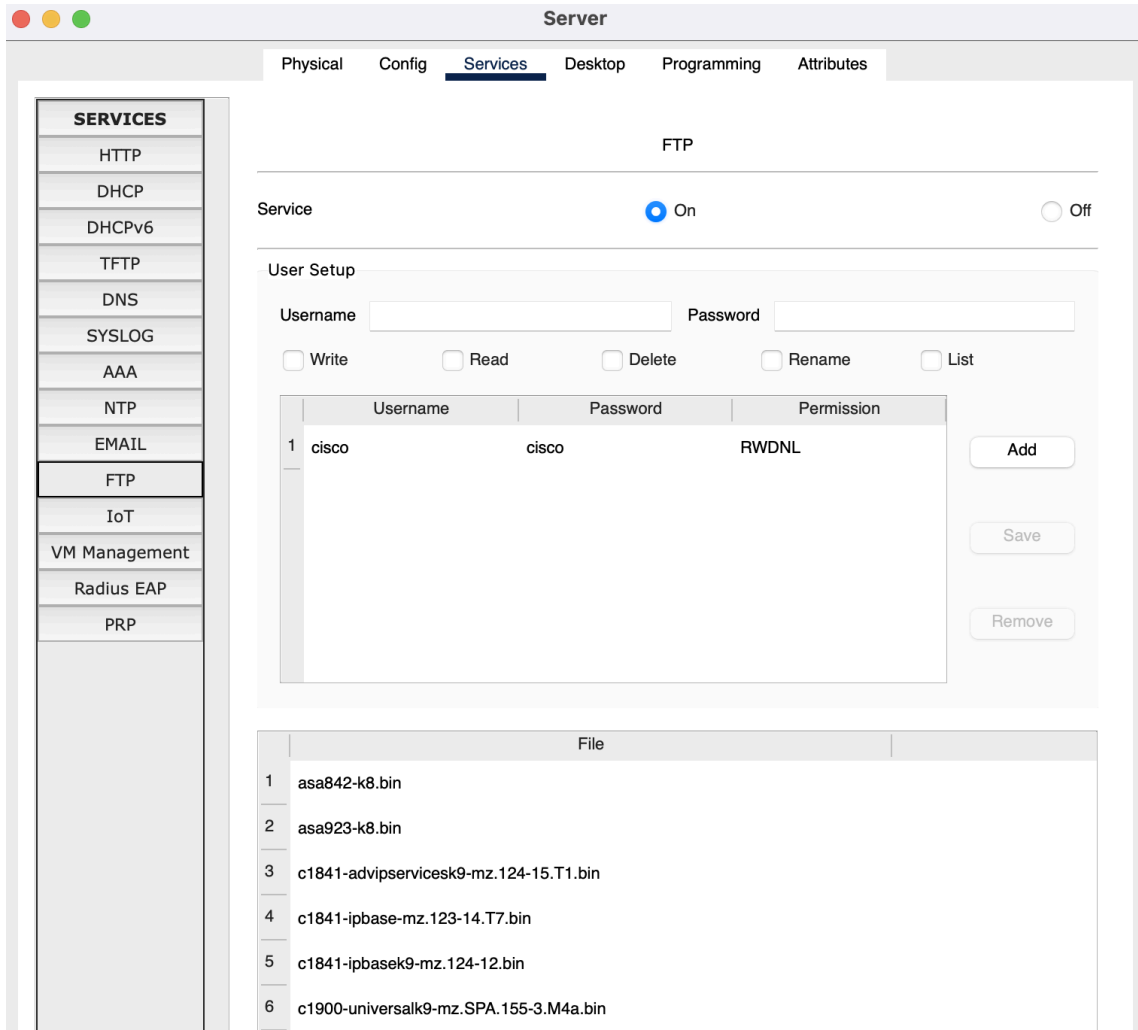
The figure shows a 'Server' configuration window with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Services' tab is active, showing a list of services on the left and configuration options for HTTP and HTTPS on the right. Below these, a 'File Manager' table is displayed.

SERVICES			
HTTP			
DHCP			
DHCPv6			
TFTP			
DNS			
SYSLOG			
AAA			
NTP			
EMAIL			
FTP			
IoT			
VM Management			

HTTP: ☒ On ☐ Off

HTTPS: ☒ On ☐ Off

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)



Since Cisco Packet Tracer does not include an nmap tool, we simulated port scanning by using Telnet to probe its open ports. We first verified which services were running on the server by checking the Services tab. HTTP (port 80) and HTTPS (port 443) were confirmed as active, as well as FTP (port 21). These represent the open ports that nmap would detect in a real environment.

Step 9: Simulate Port Scan with Telnet

The screenshot displays a Cisco Packet Tracer simulation environment. The network topology includes an ISR-331 Router connected to a 2950-24T Switch, which is connected to three PC-PT devices (PC1, PC2, PC3) and a Server-PT Server. The simulation is running, and the 'Simulation Panel' is open, showing the 'Event List'.

The 'Event List' table shows the following events:

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC2	TELNET
	0.000	--	PC2	TELNET
	0.001	PC2	Switch	TELNET
	0.002	Switch	PC1	TELNET
	0.002	Switch	PC3	TELNET
	0.002	Switch	Server	TELNET
	0.002	Switch	Router	TELNET
	0.056	--	Server	TCP
	0.057	Server	Switch	TCP
	0.058	Switch	PC2	TCP
	0.058	--	PC2	TELNET
	0.059	PC2	Switch	TELNET
	0.060	Switch	Server	TELNET
	0.114	--	Server	TCP
	0.115	Server	Switch	TCP
Visible	0.116	Switch	PC2	TCP

The 'Command Prompt' window on PC2 shows the following commands and output:

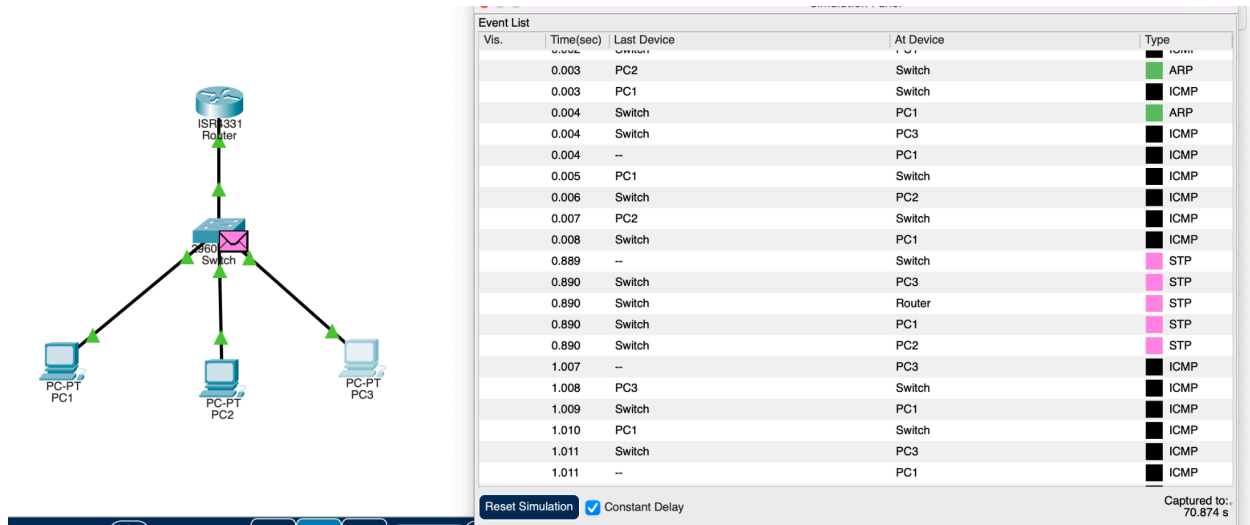
```

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.50 80
Trying 192.168.1.50 ...Open
[Connection to 192.168.1.50 closed by foreign host]
C:\>telnet 192.168.1.50 21
Trying 192.168.1.50 ...Open
  
```

We set the Simulation Mode filters to TCP and Telnet only. From PC2, we ran telnet 192.168.1.50 80 to probe port 80, then telnet 192.168.1.50 21 to probe port 21. Each Telnet connection attempt generated TCP handshake

traffic visible in the Event List. This sent TCP SYN packets to multiple ports in sequence to identify which services are available.

Step 10: Create High Volume Traffic from Multiple Sources



We ran a ping from PC1 targeting PC2, and simultaneously ran a ping from PC3 targeting PC1. This created traffic from two different sources at the same time. The Event List showed a mix of ICMP, ARP, and STP packets arriving from multiple devices simultaneously, showing high volume anomalous traffic behavior.

Results and Observations

All four tasks were completed successfully. The network was built and configured correctly with all devices communicating at 0% packet loss prior to simulation. For Task 1, the PDU Information window confirmed that packet captures in simulation mode expose key network identifiers including the source MAC address, source IP, target IP, and the ARP protocol type.

For Task 2, the ping flood simulation generated a high volume of ICMP packets from PC3 in rapid succession. The event list showed packets repeating at consistent intervals, which is consistent with DoS flood behavior. A real network would experience degraded performance or complete unavailability of the targeted host under these conditions.

For Task 3, the port scanning simulation using telnet successfully demonstrated how a scanner probes open ports. The event list showed TCP and Telnet packets being sent from PC2 to the server on different port numbers. The server responded to active

services confirming those ports are open. This is the same behavior nmap produces during a SYN scan.

Lastly for Task 4, running simultaneous pings from two different PCs produced a mixed protocol, high volume event list showing ICMP, ARP, and STP traffic from multiple sources at the same time. This pattern represents a traffic anomaly that would stand out in network monitoring tools as unusual behavior

Analysis: Normal vs. Suspicious traffic

Normal traffic:

- Single ping with 4 packets, 0% packet loss
- ARP requests only when needed to resolve addresses
- Steady, low-volume traffic from one source at a time
- TCP connection to a single known port

Suspicious traffic:

- Rapid, repeated ICMP packets from one source (ping flood)
- Multiple Telnet connection attempts to different ports in sequence
- High-volume mixed ICMP, ARP and STP packets from multiple sources simultaneously
- TCP SYN packets sent to multiple ports probing for open services

Summary of Security Threats Identified

- Ping Flood (DoS Attack): PC3 sent a continuous stream of ICMP packets targeting PCI. The Event List showed the same packets repeating at rapid intervals, which is consistent with a denial-of-service attack. In a real network, this would overwhelm the target and cause it to become unavailable.
- Port Scanning (Reconnaissance): PC2 used Telnet to probe ports 80 and 21 on the server. Each connection attempt generated TCP handshake traffic visible in the Event List. This mirrors how tools like nmap scan a target to identify which services are running before launching an attack.
- High-Volume Multi-Source Traffic (Anomaly): Running simultaneously pings from PC1 and PC3 produced a spike of mixed ICMP, ARP and STP traffic from multiple devices at the same time. This type of pattern would stand out in a real network monitoring tool as abnormal behavior worth investigating.

Challenges Faced and Solutions

Wireshark wasn't properly working:

- This issue might have happened because of compatibility issues with our firewall, or other software incompatibilities. Therefore, we decided to use Telnet instead. This saved us time as some of us already had experience with Telnet.

We weren't able to see the packets we were looking for when first entering *Simulation Mode*:

- This happened because we initially didn't set up our filters correctly. To solve it we used the ARP and ICMP filters.

Attack pings were sometimes hard to differentiate from regular pings:

- To make it easier to identify pings we used the TCP filter in Telnet. This allowed us to eliminate unnecessary noise and focus only on relevant packets.

Lessons Learned

1. **Tool selection matters and flexibility is important:** We learned that not all tools work in every environment, and issues like compatibility or permissions can limit their use. Being flexible and switching from Wireshark to Telnet allowed us to continue the lab efficiently without getting stuck.
2. **Proper configuration is essential for accurate results:** The difficulty in seeing packets showed that even small setup details, like filters, can significantly impact outcomes. Correctly configuring filters (ARP, ICMP, TCP) is crucial to capturing and analyzing the right traffic.
3. **Understanding protocols helps interpret network behavior:** Learning the differences between ARP, ICMP, and TCP made it easier to identify what type of traffic we were observing and distinguish between normal activity and potential threats.

Real-World Application

The use of firewalls is a standard practice in almost every existing network since they help prevent unauthorized communication, and protect critical systems. Some real world scenarios where what did in this lab can be applied are:

- **Monitoring a company's network:** to prevent misuse or information breaches within a network, companies may use traffic logs like we did in this lab. To detect threats they may look for abnormal patterns, inspect packets, and differentiate between normal and suspicious traffic.

- **Web server protection:** every open port represents a potential entry point for attackers; therefore it is important to verify which ports are open in a server (e.g. HTTP “port 80” / HTTPS “port 443”). In a server unused ports should be blocked to reduce attack surface.
- **Prevent university network attacks:** university networks connect thousands of students, faculty, and others at the same time, making them more vulnerable to cyberattacks. By analyzing traffic and identifying patterns like high-volume requests or repeated attempts to access different ports, IT teams can detect problems such as cyberattacks, hacked student devices, or network misuse.

Conclusion

In this lab, we successfully built and configured a network in Cisco Packet Tracer and used simulation mode to capture and analyze network traffic across four security scenarios. We captured packet details including MAC addresses, IP addresses, and protocols through the PDU information window. We simulated a ping flood attack and observed how high volume ICMP traffic from an attacker PC creates a recognizable DoS pattern. We demonstrated port scanning behavior by using Telnet to probe open services on a server, generating TCP traffic equivalent to what nmap would produce. Finally, we identified traffic anomalies by creating multi source traffic and observing the resulting mixed protocol spike in the event list.

This lab reinforced the importance of network monitoring and the ability to distinguish normal traffic from suspicious activity. Understanding what attack traffic looks like at the packet level is crucial to building effective network defenses in real-world environments.